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CE364: CRYPTOGRAPHY & NETWORK SECURITY
Practice Questions

	Caesar Cipher/Additive Cipher
1	Encrypt the message using shift key k = 3: **ATTACK AT DAWN**
2	Ciphertext: **UFTU NFTTBHF** 1. Try all possible shifts. 2. Identify the correct plaintext. **3.** How many possible keys exist in Caesar cipher?
3	During a cybersecurity monitoring operation, a security analyst intercepts the following encrypted message from an unknown attacker group: **FYYFHP FY IFBS** The analyst suspects that the attacker used a Caesar Cipher to encrypt the original message. Further investigation reveals that the encryption key used was k = 5. 1. Decrypt the message step by step.
	Multiplicative Cipher
4	During a classified communication, the plaintext: **SAFE COMMUNICATION** is encrypted using a Multiplicative Cipher with key = 9. 1. Encrypt the message.
5	A cybersecurity analyst intercepts the following encrypted message: **UZZUYM UZ JUEH** It is known that: A Multiplicative Cipher was used. The secret key used was k = 5.
	Affine Cipher
6	A cybersecurity analyst intercepts the encrypted message: **IHHWVCSWFRCP** Intelligence reports confirm that an Affine Cipher was used with: $a = 5, b = 8$ The analyst must recover the original plaintext.
7	Plaintext: **SECURITY** Keys:

	$\mathbf{a = 7, b = 3}$ Using: A = 0, B = 1, …, Z = 25.Use Affine cipher to encrypt the message
	Playfair Cipher
8	During a high-level diplomatic mission, an intelligence officer must send a confidential message to headquarters. To avoid interception through frequency analysis attacks, the officer chooses the Playfair Cipher encryption technique. The keyword selected for generating the matrix is: **CYBERSECURITY** The plaintext message is: **DEFEND THE NETWORK** (Spaces removed before encryption.)
9	During a cyber-defense operation, an intelligence unit must securely transmit a classified alphanumeric message containing both letters and numbers. The keyword selected for matrix construction is: **SECURE2026** Plaintext Message **ATTACK AT 2100 HOURS** (Remove spaces before encryption.)
10	During a classified cyber-intelligence operation, a security analyst intercepts the following encrypted alphanumeric message transmitted over a secure military channel: **R8Y$ ZK1@ QP7L X3#F** The keyword used to construct the 7×7 matrix was: {CYBERSECURITY2026@#} • Symbols: @ # $ % & * + - = ? ! _ . The analyst must now decrypt the message.
	Hill Cipher
11	An analyst encrypts the message: **SAFE** Using the key matrix: $\boldsymbol{K} = \begin{bmatrix} \mathbf{11} & \mathbf{8} \\ \mathbf{3} & \mathbf{7} \end{bmatrix}$
12	During a high-priority cybersecurity operation, a defense unit must securely transmit a classified message to headquarters. To strengthen encryption beyond classical substitution methods, the unit uses a Hill Cipher with a 3×3 key matrix. The selected key matrix is: $\boldsymbol{K} = \begin{bmatrix} \mathbf{6} & \mathbf{24} & \mathbf{1} \\ \mathbf{13} & \mathbf{16} & \mathbf{10} \\ \mathbf{20} & \mathbf{17} & \mathbf{15} \end{bmatrix}$ The plaintext message is: **ACTNOW** (Use A = 0, B = 1, …, Z = 25)
13	During a classified satellite communication mission, a defense organization needs to encrypt a sensitive command before transmission.

	To enhance cryptographic strength and resist frequency analysis, the organization uses a **Hill Cipher with a 4×4 key matrix**. The selected key matrix is: $$K = \begin{bmatrix} 3 & 10 & 20 & 5 \\ 20 & 9 & 17 & 8 \\ 9 & 4 & 17 & 6 \\ 15 & 14 & 2 & 11 \end{bmatrix}$$ The plaintext message is: **DEFENDTHEBASE** (Use A = 0, B = 1, …, Z = 25)
	Polyalphabetic Cipher
14	An intelligence officer encrypts the message: **NETWORK SECURITY** Using the keyword: **SAFE**
	Vigenere Cipher
15	During a cybersecurity monitoring operation, a classified message must be encrypted before transmission. To resist simple frequency analysis, the analyst uses a **Vigenère Cipher** The selected keyword is: **NETWORK** The plaintext message is: **INFORMATION SECURITY**
	Transposition Cipher
16	During a cyber-defense operation, a classified message must be securely transmitted to headquarters. Instead of substituting letters, the analyst chooses a **Columnar Transposition Cipher** to rearrange the message structure. The keyword selected for encryption is: **CIPHER** The plaintext message is: **DEFEND THE NETWORK**
17	An intercepted ciphertext reads: **TKAATNACDTAW** It is known that: • A **Columnar Transposition Cipher** was used. • The keyword was: **SAFE**